

MURREE, PakistanWMO: **415730**Lat: **33.917N**Lon: **73.383E**Elev: **2127**StdP: **78.25**Time Zone: **5.00 (E05)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	-0.9	-12.7	1.6	8.2	-10.6	2.0	8.3	6.5	4.6	5.3	5.4	1.1	0	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	7.6	28.0	16.7	26.5	17.0	25.2	17.1	21.5	23.5	21.0	22.9	20.4	22.4	1.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	20.3	22.8	20.4	19.6	22.3	19.8	18.9	21.8	75.0	23.4	72.7	22.9	70.3	22.6	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.2	4.0	3.2	-3.7	30.5	2.0	1.9	-5.1	31.9	-6.3	33.0	-7.4	34.0	-8.8	35.4		
(5)			-5.3	22.9	1.9	0.4	-6.6	23.1	-7.7	23.4	-8.8	23.6	-10.2	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	5.1	5.6	10.3	14.8	18.9	20.8	20.2	19.5	18.4	15.8	11.8	8.1
	DBStd	6.21	3.17	3.22	4.40	3.71	3.01	2.60	1.64	1.17	1.60	2.39	2.61	3.36
	HDD10.0	427	153	126	52	6	0	0	0	0	0	1	13	77
	HDD18.3	1788	409	355	251	117	30	6	2	3	17	83	197	317
	CDD10.0	1943	2	4	61	148	274	326	316	296	253	180	66	18
	CDD18.3	262	0	0	1	10	46	82	59	41	20	4	0	0
	CDH23.3	844	0	0	1	25	214	438	131	19	14	2	0	0
	CDH26.7	111	0	0	0	1	26	71	14	0	0	0	0	0
Wind	WSAvg	1.1	1.2	1.3	1.4	1.4	1.3	1.1	0.7	0.6	0.9	1.0	1.0	1.1
Precipitation	PrecAvg	1752	131	146	143	130	89	138	337	311	163	63	32	55
	PrecMax	2305	295	345	244	389	208	519	566	459	314	255	104	192
	PrecMin	1284	0	7	16	24	24	37	163	151	23	0	0	0
	PrecStd	299	97	74	72	88	44	115	99	77	90	69	32	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	16.0	22.1	25.7	28.9	30.3	28.4	24.9	24.7	23.4	20.2	18.5
		MCWB	5.6	8.2	13.2	15.7	16.1	17.6	18.5	20.5	18.3	16.2	11.0	8.9
	2%	DB	12.9	14.0	19.8	23.7	27.2	28.6	26.1	23.8	23.6	22.1	18.5	16.0
		MCWB	5.0	6.5	11.9	14.1	15.5	16.7	19.5	20.3	18.3	15.5	10.1	7.5
	5%	DB	11.4	12.1	18.2	22.1	25.6	27.2	24.8	23.1	22.7	20.9	17.1	14.3
		MCWB	4.5	5.6	10.5	13.4	15.2	17.1	19.5	20.1	18.1	14.5	9.5	6.4
	10%	DB	9.8	10.6	16.5	20.4	24.3	25.9	23.6	22.4	21.8	19.7	15.8	12.7
		MCWB	3.6	4.8	9.5	12.3	14.5	17.1	19.5	19.9	18.0	14.0	8.8	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	9.6	14.1	16.6	18.2	21.8	22.3	22.2	20.8	18.7	12.6	10.4
		MCDB	12.4	14.6	21.1	23.7	25.4	24.9	24.3	23.7	22.8	21.6	18.3	16.9
	2%	WB	5.9	7.8	12.7	15.2	17.2	20.8	21.6	21.4	20.0	17.5	11.4	8.4
		MCDB	11.1	12.5	19.1	22.1	24.6	24.2	23.6	22.9	22.2	20.5	17.0	14.3
	5%	WB	5.1	6.3	11.5	14.1	16.1	19.9	21.0	20.9	19.3	16.2	10.5	7.3
		MCDB	9.9	11.0	17.4	21.0	23.9	23.6	23.0	22.4	21.6	19.4	16.1	12.9
	10%	WB	4.3	5.3	10.1	13.1	15.3	18.9	20.4	20.3	18.6	15.0	9.4	6.2
		MCDB	9.0	9.5	15.7	19.5	23.1	23.1	22.5	21.9	21.0	18.8	14.8	11.5
Mean Daily Temperature Range	5% DB	MDBR	7.0	6.5	7.0	7.5	7.9	7.6	5.5	5.3	6.5	7.5	7.4	7.3
		MCDBR	8.7	8.4	8.6	9.0	9.1	8.9	7.8	6.8	7.7	8.3	8.6	8.9
		MCWBR	5.7	5.9	6.1	6.6	6.3	6.4	5.7	5.9	6.7	6.7	6.6	6.1
	5% WB	MCDBR	8.0	7.7	8.4	8.7	8.7	6.8	5.8	5.5	6.7	7.2	8.0	8.1
		MCWBR	5.8	5.9	6.3	6.6	6.6	6.0	5.2	5.2	6.2	6.3	6.4	6.2
Clear-Sky Solar Irradiance	taub	0.319	0.357	0.366	0.380	0.396	0.448	0.495	0.474	0.403	0.367	0.342	0.316	
	taud	2.292	2.211	2.237	2.242	2.209	2.071	1.984	2.053	2.223	2.235	2.241	2.313	
	Ebn at Noon	882	881	905	906	891	841	798	810	856	858	844	862	
	Edh at Noon	103	125	132	138	144	165	179	165	133	120	107	95	
All-Sky Solar Radiation	RadAvg	2.83	3.34	4.62	5.77	6.87	6.96	6.02	5.47	5.35	4.66	3.43	2.77	
	RadStd	0.34	0.52	0.50	0.39	0.41	0.37	0.48	0.43	0.34	0.27	0.24	0.23	